1P STUDIO

FirstPass STUDIO

USER GUIDE

Broadcast graphics for netball livestreams, fed live from PlayHQ and run from one Mac.

VERSION 1 · OCTOBER 2026 · BY PIVT

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● LIVE FROM PLAYHQ



USER GUIDE

Graphics that follow the game.

Broadcast graphics for netball livestreams, fed live from PlayHQ and run from one Mac.

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ABOUT THIS GUIDE

Graphics that follow the game.

FirstPass Studio is a Mac app that draws broadcast graphics for a netball livestream and sends them to your vision switcher. The scores, clocks, ladders, fixtures and line-ups come from PlayHQ's public data, so there is nothing to type in during the game.

This guide starts with the way most nights run, one operator and Studio doing the rest, then covers running it by hand, setting up a competition, sponsors, athletes and the Stream Deck.

5 s
The live score, quarter and clock read from PlayHQ every 5 seconds while the game is scored.

31
Broadcast graphics, from the scoreboard and ladder to the sponsor wall, plus a test pattern.

17
Competition logo animations, and six intro and outro styles.

1
Operator on a followed night. Sequences run the run-up, the breaks and full time.

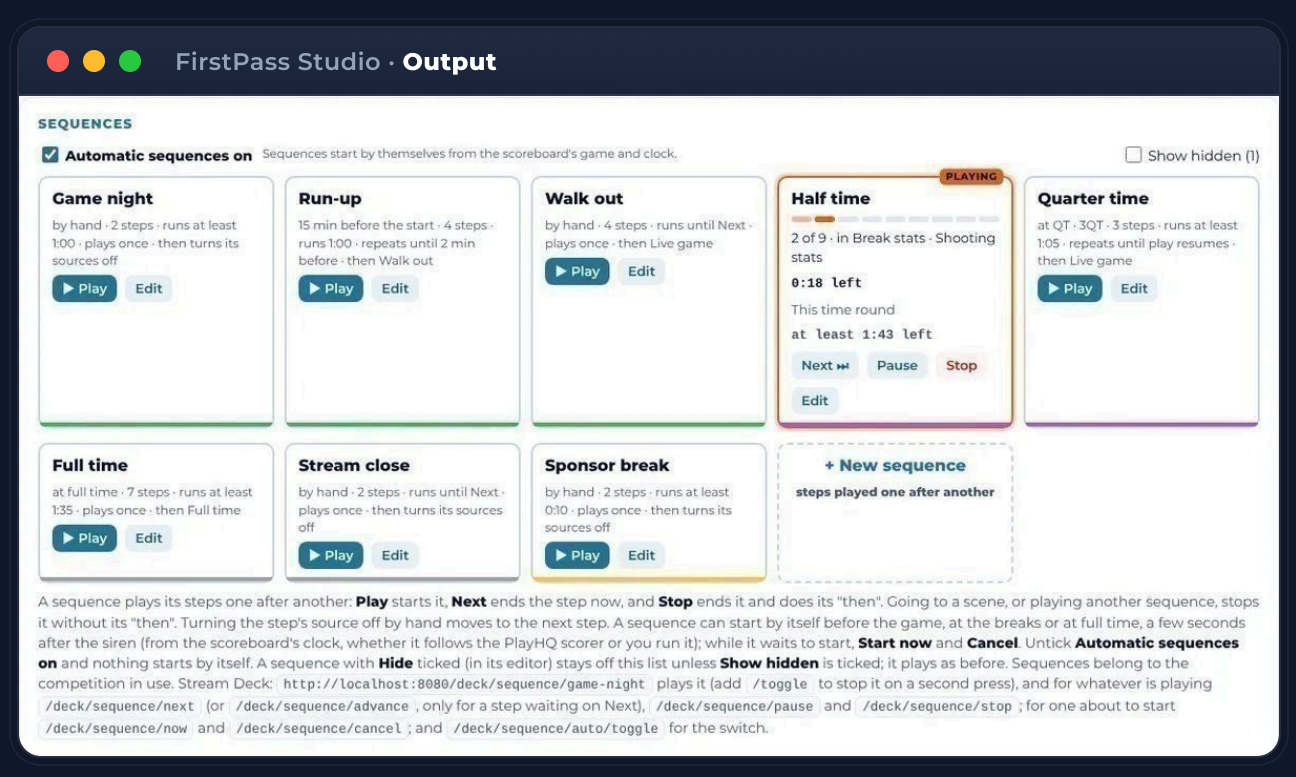
IN
PlayHQ
The fixture, ladder, line-ups and the live score from the scorer's tablet. No account, no key.

→

ON YOUR MAC
FirstPass Studio
Builds one 1920 × 1080 scene from layers: the base scene, automatic layers, scenes and sequences.

→

OUT
Your switcher
A display, a Blackmagic device or NDI, with fill and key or a chroma colour over the cameras.





The Output page runs the show; the super shot plays by itself.

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Every screenshot comes from Studio running the fictional Elite Netball League: eight clubs named after animals, three divisions, a live game and fictional sponsors.

CHAPTER 1 OF 11

What Studio is

01

FirstPass Studio is a Mac app that draws broadcast graphics for a netball livestream and sends them to your vision switcher. The scores, clocks, ladders, fixtures and line-ups come from PlayHQ's public data, so there is nothing to type in during the game.

1 The words used in this guide
2 The pages you work in
3 The sample competition in this guide

Studio does one job on stream night: graphics. Your cameras, switcher and streaming encoder stay as they are. Studio hands the switcher a picture with transparency (fill and key, or a chroma colour) so the graphics sit over the vision.

1.1 The words used in this guide

Studio

The whole thing: the Mac app, the pages you work in, and the graphics it puts on screen.

On air, off air

Studio is on air while a Studio window is open. On air it talks to PlayHQ, serves the graphics and answers the Stream Deck. Closing the last window takes it off air.

A graphic, or a page

One thing that can go on screen: the scoreboard, a ladder, a lower third, an animation. Each is a web page that Studio draws.

A source

A graphic placed in the scene, with its own position, options and fade. The scoreboard is one source; a ladder is another.

The scene

Studio's own 1920 × 1080 frame, built from sources switched on and off. It is what the switcher receives.

A scene (on the Scenes list)

A saved look: which sources are on and which are off. "Live game" and "Clear" are scenes.

A sequence

A list of steps played in order from one press: a run-up, a quarter-time break, a full-time wrap.

The competition

Everything that belongs to one league: its PlayHQ link, names, rules, colours, logos, sponsors and athletes. Studio keeps a setup per competition and uses one at a time.

The data folder

Where Studio saves everything, outside the app, so updates never touch it.

1.2 The pages you work in

Every Studio window has the same tab bar: **Output**, **Production Planner**, **Scoreboard control**, **Competitions setup**, **Sponsors**, **Athletes**, **Stream Deck**, **Socials**, **Recordings** and **Docs** (⌘1 to ⌘9, then ⌘0, in the app). Output is where the show is built and sent out; the Production Planner plans the court and the commentators for each night; Scoreboard control runs the board; Socials makes the night's pictures for social media; Recordings keeps each night on air; the rest are set up before the night.

> *Studio does one job on stream night: graphics.*

1.3 The sample competition in this guide

The screenshots use a fictional league, the **Elite Netball League**, with eight clubs named after animals (Falcons, Panthers, Stingrays, Cobras, Grizzlies, Hawks, Rhinos and Wolves) across Premier, Men's and Under 21 divisions, playing at the Metro Arena. Its sponsors are fictional too. Everything you see is what Studio draws for any competition once its setup is in.



CHAPTER 2 OF 11

Getting started

Install the app, import or make your competition, and learn the one rule that matters on stream night: keep a Studio window open.

1 Install · 2 On air and off air · 3 The 1P menu · 4 Before you go on air

2.1 Install

01
Download `FirstPass Studio.dmg`, drag the app to Applications and open it. It runs on Apple silicon Macs (M1 and later).

02
The **1P** icon appears in the menu bar, off air. Choose **Open Studio**. A short splash shows, Studio goes on air and the Output page opens in the app's own window.

03
Go to **Competitions setup** and either **Import a template file** (your league's `.json`, with its logos and sponsors inside) or **New competition** and fill it in (Chapter 6).

04
Back on **Output**, add the graphics you need with **Add page**, and turn on the video output that feeds your switcher (Chapter 3).

2.2 On air and off air

Studio only runs while a Studio window is open. Opening one (Open Studio, any page in the 1P menu, or ⌘N) puts it on air; closing the last one takes it off air, and the graphics and Stream Deck get no answer until a window opens again. Closing the last window while on air asks first, and so does ⌘Q, so a stray keypress can't stop a stream. Leave the Output page open all night.

The status line under the Output page's title shows the game on the board, whether scores and clock follow PlayHQ, how long since PlayHQ last answered (amber when overdue, red after a minute of errors) and whether the video output is on. If anything is wrong, this line says so on the page you keep open.

2.3 The 1P menu

Open Studio, Scoreboard control, Competition setup, Copy the Studio link, Devices..., Install the Stream Deck plugin, Back up... and Restore..., Settings folder (iCloud Drive), Open the settings folder, Open the debug log, Open the app log, Restart Studio, Open at login, Only this Mac can change things, What's new..., Check for updates, Stop Studio (while on air) and Quit. Updates download in the background and install when the app quits, never mid-stream; **Restart to update** installs one straight away.

LINKS Studio's links start with the Mac's Bonjour name, such as `http://matts-macbook-pro.local:8080/`, so the same link works from a phone, another computer or a Stream Deck on the same network. `http://localhost:8080` works on the Mac itself.

> **The one rule that matters on stream night: keep a Studio window open.**
>
> CHAPTER 2 · GETTING STARTED

2.4 Before you go on air

01
Open the app. The Output page opens and Studio goes on air. Keep a window open all night.

02
Right competition. Competitions setup shows the one in use at the top. Check its PlayHQ link, the court to follow and the scoring window.

03
Right game. On Scoreboard control, the picked game is tonight's first on your court and Follow PlayHQ is on.

04
Scores arrive. Once the scorer has started on PlayHQ, the control page says the game is being live scored and the board moves. Nothing is read before the scoring window opens, so a quiet board before then is normal.

05
Scene and output. On Output, turn on the display, NDI or Blackmagic output your switcher takes, check fill and key or chroma, and show one graphic to see it land.

06
Sound. Play a few seconds of music and an animation with sound, and check the sound only comes out where you want it.

07
Stream Deck. Press a Source key and watch the Sources list follow it. A score key switches scoring to Manual, so put it back to Follow PlayHQ after testing.

08
Sponsors. Tonight's spots and ads are on; the sponsor report starts counting when the output turns on.

Game night, the automated way

03

Once a three-night fixture can be scheduled, the next step is to make sure the night itself runs without anyone having to think about it. Pinellas Cricket's **Game night** feature is the automation layer that takes over from the moment the first ball is bowled.

Auto Scoreboard · Broadcast Mode · Live Graphics · Walkout Music · Scoreboard Control · Audio Mixer · Scene Switcher · Video Overlays · Timed Graphics · Tournament Mode · Replay · Recording

How it fits together

- **Video Playlist.** The playlist loads the Scene Switcher, audio mixer and overlay templates for the night's broadcast. Operators can swap templates mid-game without losing the playlist.
- **Broadcast Mode.** Switches the interface into a clean, full-screen layout for the control room, hiding editing tools and showing only the live feed and essential controls.
- **Automatic layers.** Handles the play-by-play data and other graphics, updating them in real time as the match progresses.
- **Scoreboard control.** Connects to the scoring system and keeps the on-screen scoreboard in sync with the live match.
- **Tournament.** Links the night to the wider tournament, so results and standings update automatically when the game finishes.



> **A sequence can start by itself from the game's clock, carry music, and hand over to another sequence.**

Scoreboard control



Scoreboard control

Scoreboard control connects to the scoring system and keeps the on-screen scoreboard in sync with the live match. Operators can choose which fields to display, such as batting team, bowling team, current over, and run rate, and the overlay updates as the game progresses.

Auto Scoreboard builds the scoreboard overlay automatically from the match data, so it appears as soon as the game starts and updates after every ball. Operators can override any value manually if the feed is wrong.

Live Graphics lets you add custom graphics on top of the scoreboard, such as player milestones or partnership totals, and position them anywhere on the screen.

Timed Graphics schedules graphics to appear and disappear at set times or after a set number of overs, so they don't need to be managed by hand.

Loaders & Lower 3rd



Loaders & Lower 3rd

Loaders & Lower 3rd is the place to build and edit the animated graphics that appear over the video feed. Players can design lower thirds for batters, bowlers, and umpires, as well as transitional loaders that play between scenes. Templates can be saved and reused across matches.

Walkout Music plays the right track as each batter comes to the crease. Operators can assign tracks to individual players in advance, or change them on the fly during the match.

Video Overlays adds persistent graphics on top of the video, such as a channel bug, sponsor boards, or a match clock. Overlays can be toggled on and off independently of the scene.

Audio Mixer gives you control over the levels of every audio source in the broadcast, including commentary, crowd, and walkout music. Operators can mute, balance, and route audio to different outputs.

Scene Switcher lets you cut between cameras and graphics with a single click or tap. Scenes can be arranged in any order and assigned to hotkeys for fast switching during play.

Replay captures the last few moments of play and plays them back on demand, with adjustable speed and the ability to mark in and out points.

Recording records the broadcast to disk, with options for quality and file format. Recordings can be started and stopped manually or scheduled to run automatically.

The Output page

The Output page builds the broadcast scene layout, from the camera feeds to the graphics overlays. Each **Video Output** is a separate canvas, and you can have as many as you need for different screens or streams.



The Output page

Each output has a **Preview** and a **Program** monitor, so you can line up the next shot before it goes live. Add **Sources** for cameras, graphics, and audio, then arrange them in the **Scene Switcher** with cuts, fades, or stingers between them.

Use **Hotkeys** to trigger scenes, replays, and recordings without taking your hands off the match. Set up **Replay** to mark in and out points on the fly, and schedule **Recording** to start and stop automatically at the right moments.

The base scene



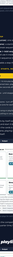
The base scene

The **base scene** is the foundation of every broadcast. It's the layer that sits behind all the graphics and overlays, and it's usually a live camera feed or a static image.

- **Scoreboard.** Turns it on and off, and lets you choose which scoreboard style to display.
- **Live Graphics.** Adds and removes graphics on top of the base scene, like lower thirds or player cards.
- **Timed Graphics.** Schedules graphics to appear and disappear at set times or after a set number of overs.
- **Tournament.** Pulls in fixture and standings data so you can show what's coming up next.

Pick the base scene from the **Scene Switcher** on the Output page. You can change it at any point during the match without losing your overlays.

Automatic layers

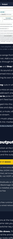


Automatic layers

Automatic layers are driven by the fixture's play-by-play data, so they update on their own as the match progresses. There's no need to touch them once they're set up.

- **Auto Scoreboard.** Keeps the scoreboard overlay in sync with the live score, wickets, and overs.
- **Broadcast Mode.** Switches the interface into a clean, full-screen layout for the control room, hiding editing tools and showing only the live feed and essential controls.

Scenes



Scenes

Scenes are the building blocks of your broadcast. Each one is a saved layout that combines a camera feed, graphics, and audio into a single look you can switch to with a click or a hotkey.

Create as many scenes as you need for the different moments in the game, such as a wide shot for the start of play, a close-up for the batter, or a full-screen graphic for the scoreboard. Give each scene a clear name so you can find it quickly during a busy over.

Sequences



Sequences

Sequences automate a series of steps, such as a graphic appearing, a camera cutting in, and music starting, all at the right moment. They can be triggered by hand or by match events, so the broadcast runs itself.

Build a sequence by adding steps in order and setting how long each one lasts. You can also add delays between steps to time things to the action on screen.

THREE KINDS OF STEP

- **Graphics.** Add lower thirds, scoreboards, and other overlays to the broadcast.
- **Scene.** Switch to a different camera angle or graphic layout.
- **Audio.** Start walkout music, play a sting, or duck the crowd mic.

WHEN A SEQUENCE STARTS, BEFORE THE BALL

Manual	Start it yourself from the Sequences panel.
Batter walks out	Fires when a new batter comes to the crease.
Wicket	Fires when a wicket is recorded.
Over	Fires when the over reaches a set ball. Choose which ball.
Innings	Fires when the innings reaches a set ball. Choose which ball.

WHEN A SEQUENCE ENDS, BEFORE THE BALL

Manual	Stop it yourself from the Sequences panel.
Batter walks out	Stops when the batter reaches the crease.
Wicket	Stops when a wicket is recorded.
Over	Stops when the over reaches a set ball. Choose which ball.
Innings	Stops when the innings reaches a set ball. Choose which ball.



Sequences

Sequences live on the Scenes panel, just under the **Live Graphics** section. Click **New Sequence** to create one, then add steps and set when it starts and ends.

Each step can be a graphic, a scene change, or an audio cue, and you can drag them into the order you want. Preview a sequence before you go live to make sure the timing feels right.

Music, playlists and mics



Music, playlists and mics

Pinellas Cricket has three ways to play music during a broadcast: **Walkout Music** for batters coming to the crease, **Playlists** for background tracks, and **Mics** for commentary and crowd audio.

Walkout Music is set up per player, so each batter gets their own track when they walk out. Playlists run in order or on shuffle, and you can skip tracks or adjust the volume without stopping the broadcast. Mics let you route microphones to the right outputs and control their levels on the fly.

WALKOUT MUSIC

Walkout Music plays a player's track when they walk out to bat. Assign tracks from the player's profile before the match, or change them during play if a batter wants a different song.

- **Start offset.** Skip the intro and start the track at the chorus.
- **Fade in.** Ease the track in over a few seconds instead of starting at full volume.
- **Duck commentary.** Lower the track automatically when the commentary mic is live.

PLAYLISTS

Playlists run in the background during breaks, drinks, and between innings. Add tracks from your library, drag them into order, or switch to shuffle for a mixed set.

- **Crossfade.** Overlap the end of one track with the start of the next for a smooth transition.
- **Volume.** Set the playlist level independently of walkout music and commentary.
- **Skip.** Move to the next track without stopping the broadcast.

MICS

Mics handles commentary and crowd audio. Route each microphone to the outputs you need, and set its level so it sits well in the mix.

- **Commentary.** Send the commentary mic to the broadcast and stream outputs.
- **Crowd.** Add ambient crowd audio to the broadcast for atmosphere.
- **Mute.** Cut a mic instantly when you don't want it heard.

Video output

The **Video output** section of the app lets you keep an eye on the scores as you work on the broadcast. Pinellas Cricket supports three kinds of output, and you can have as many of each as you like.

	Scores	Graphics
Fullscreen	Shows the live scores on a full screen. Use this on a TV or monitor in the control room.	Shows the live scores on a full screen. Use this on a TV or monitor in the control room.
Overlay	Shows the live scores as a transparent layer you can add to your broadcast software.	Shows the live scores as a transparent layer you can add to your broadcast software.
NDI	Sends the live scores over your network using NDI.	Sends the live scores over your network using NDI.

SET UP

Open the **Video output** section and click **Add output**. Choose the type you want, then pick the screen or network interface it should use. Give it a name you'll recognise later, like "Control room TV" or "Stream overlay".

WHAT TO SHOW ON EACH OUTPUT

Each output can show the scores, the graphics, or both. Use **Scores only** for a clean feed you'll add graphics to yourself, **Graphics only** for a transparent overlay, or **Scores and graphics** for a ready-to-use broadcast feed.

HOW TO CONTROL THE OUTPUT

Use the **Start** and **Stop** buttons to turn an output on or off. You can also set it to start automatically when you open the app, so it's ready before the match begins.

WHAT YOU'LL SEE ON SCREEN

The scores and graphics on your output update in real time as the match progresses. You'll see the batting team's total, overs, and run rate, plus any graphics you've added to the broadcast.

TIPS

- **Check your output is working before the match starts.** Open it on the screen you'll be using and make sure the scores are showing correctly.
- **Use Overlay when you're adding graphics in another app.** It keeps the background transparent so you can layer it over your camera feed.
- **Turn off outputs you're not using.** Each one takes a little extra processing power, so stopping the ones you don't need can help the app run smoothly.

Tournament

Tournament is the hub for everything that sits above a single match. It's where you create the competition, add the teams, and keep the standings up to date as results come in.



Tournament

SETTING UP A TOURNAMENT

Click **New tournament** and give it a name, a start date, and the number of overs per match. You can add teams one by one or import a full squad list from a spreadsheet.

FIXTURES AND RESULTS

Fixtures are generated automatically once you've added your teams. Choose between a round-robin, knockout, or group-stage format, and edit any match if you need to change the date or venue.

Results are entered after each game, or pulled in automatically if you're using the scoring system. Points tables update as soon as a result is confirmed.

STANDINGS AND STATS

The **Standings** tab shows the current points table, net run rate, and qualification status. Use the **Stats** tab to see top batters, bowlers, and team records for the tournament so far.

SHARING WITH PLAYERS

Click **Share** to publish a read-only link to the fixtures, results, and standings. Players and supporters can follow along without needing an account.

A night, start to finish

Here's how a typical night looks, from opening the playlist to the night's recordings.

Evening

Arrive
Open the app and check the night's fixture.

Audio and Mixer
Set the playlist, mics, and levels for the night.

Video Output
Add a Scores only output for the control room TV.

Add a Graphics only output for the stream.

Tournament
Confirm the fixture and check the standings.

Game night
Start the night's broadcast.

Scene Switcher
Set the opening scene and hotkeys.

Sequences
Start the pre-match sequence.

Walkout Music
Assign tracks to the batters.

Match starts
Switch to Broadcast Mode.

Live Graphics
Add lower thirds and player cards as the game goes on.

Timed Graphics
Schedule graphics for drinks breaks and between innings.

Replay
Mark in and out points on the fly.

Recording
Start recording when the first ball is bowled.

Match ends
Stop the recording and review the footage.

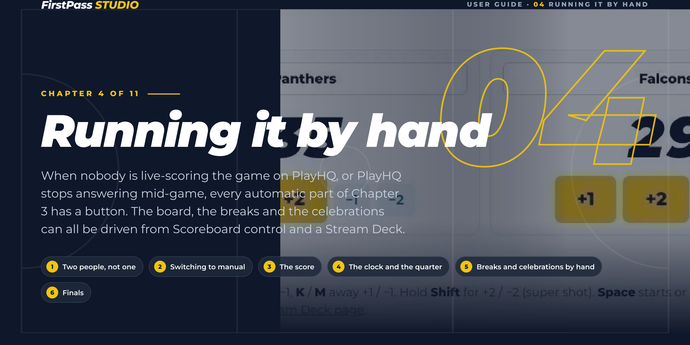
The Production Planner

Production Planner gives you the space to plan every detail of your broadcast before the match starts. Lay out your scenes, graphics, and audio cues in advance so you're ready to go live with confidence.



Production Planner

- **Production Planner.** Plan your broadcast ahead of time, including scenes, graphics, and audio cues.
- **Scene Switcher.** Switch between cameras and graphics with a single click or tap.
- **Video Overlays.** Add persistent graphics on top of the video, such as a channel bug, sponsor boards, or a match clock.
- **Live Graphics.** Add and remove graphics on top of the base scene, like lower thirds or player cards.
- **Timed Graphics.** Schedule graphics to appear and disappear at set times or after a set number of overs.
- **Audio Mixer.** Control the levels of every audio source in the broadcast, including commentary, crowd, and walkout music.
- **Walkout Music.** Play the right track as each batter comes to the crease.
- **Tournament.** Pull in fixture and standings data so you can show what's coming up next.
- **Replay.** Capture the last few moments of play and play them back on demand.
- **Recording.** Record the broadcast to disk, with options for quality and file format.



CHAPTER 4 OF 11

Running it by hand

When nobody is live-scoring the game on PlayHQ, or PlayHQ stops answering mid-game, every automatic part of Chapter 3 has a button. The board, the breaks and the celebrations can all be driven from Scoreboard control and a Stream Deck.

1 Two people, not one
2 Switching to manual
3 The score
4 The clock and the quarter
5 Breaks and celebrations by hand
6 Finals

Studio still needs the competition on PlayHQ for its fixture, ladder, squads and results; what changes in manual mode is who runs the score and the clock. The same competition can be followed one week and run by hand the next, and nothing on the Output page changes between the two.

4.1 Two people, not one

A followed game can be run by one person because the score, the clock and the breaks take care of themselves. A game run by hand is a different job: someone has to press a key for every goal within a second or two of it, start and stop the clock with the umpire's whistle, and still call the breaks and the graphics. We recommend two people share it:

Camera operator

- Frames and follows play on the cameras, and cuts between them on the switcher.
- Runs the game clock from the whistle, with **Start / Stop** and **Reset clock** keys beside the camera controls.
- Scores the goals as they go in, **+1** or **+2**, because they are already watching the ring.
- Nothing else: no graphics, no breaks, no music.

Production manager

- Keeps the Output page open and watches the stream itself, not the court.
- Corrects the score (**−1**, **−2**) and the quarter, and sets the clock when it drifts from the venue's.
- Plays the run-up, break and full time sequences from their keys, and the lower thirds, ladders and stats during play.
- Runs the music, the sponsor ads and the end of night, and picks the finals winners.

If one person must do it alone, keep the Stream Deck to the score and clock keys and let the sequences do the breaks, exactly as they do in automated mode: the game triggers follow the scoreboard's clock whoever runs it.

> ***If one person must do it alone, keep the Stream Deck to the score and clock keys and let the sequences do the breaks.***
>
> CHAPTER 4 · RUNNING IT BY HAND

4.2 Switching to manual

Press **Manual** on Scoreboard control, or just press any score key: either switches the board to manual scoring and stops following the PlayHQ clock as well. Press **Follow PlayHQ** to go back; the score catches up with PlayHQ on the next check and the clock follows the scorer's again.

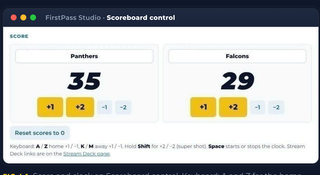


FIG 4.1 Score and clock on Scoreboard control. Keyboard: A and Z for the home team, K and M for away, Shift for a super shot, Space for the clock.

WHEN PLAYHQ DROPS OUT

If PlayHQ stops answering during a followed game, the scoreboard keeps its last score, the ladder, fixtures and game list keep what they last read, and the debug log records each failed read. Switch to Manual and run the board by hand; Follow PlayHQ picks it up again when PlayHQ is back, and corrects the score to the scorer's.

4.3 The score

Each team has **+1**, **+2**, **−1** and **−2**. A +2 is a super shot: it scores two and plays the super shot celebration, so a +2 pressed by mistake is fixed with −2 (which plays nothing). **Reset scores to 0** clears both. The hot streaks and game moments still play in manual mode, because Studio works out the goal order from the presses just as it does from PlayHQ's score.

KEY	WHAT IT DOES
Home +1 / Away +1	A goal.
Home +2 / Away +2	A super shot: two points and the celebration.
−1, −2	Take a goal or a super shot back, quietly.
Reset scores to 0	Both scores to zero, for the next game.
Super shot (home / away)	The celebration only, score unchanged: for a test, or a super shot that was scored as +1 +1.

4.4 The clock and the quarter

Start and **Stop** are one key on the Stream Deck. **Reset clock** puts a full quarter back (the competition's quarter length; a key can set any time). **+10 s** and **−10 s** nudge it, and a key can add or take off any amount. Type the venue's clock in **Match venue clock** to line the two up, and untick **Show clock** for a game with no reliable clock at all. Quarters are **Q1** to **Q4** plus extra time, or H1 and H2 for a competition played in halves; **Next quarter** steps through them. Changing the quarter doesn't touch the clock, so make the quarter key a Stream Deck Multi Action that also resets the clock.

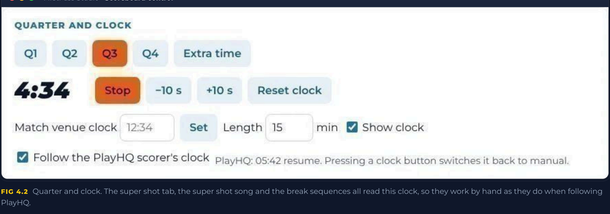


FIG 4.2 Quarter and clock. The super shot tab, the super shot song and the break sequences all read this clock, so they work by hand as they do when following PlayHQ.

4.5 Breaks and celebrations by hand

A sequence that starts by itself at quarter time is started by its key too, so the production manager can run **Quarter time** the moment the siren goes instead of waiting for the clock, or run **Half time** early when a game is called. **Next**, **Pause** and **Stop** work on whatever is playing, and **Automatic sequences on** on the Output page can be unticked for a night where every break is called by hand.

- **Winner.** One key. The celebration reads the board's score when it comes on, so it covers a home win, an away win and a draw. Check the score first. A source with **Result** set to home, away or draw forces it.
- **Super shot and streaks, once.** Besides the automatic sources, a Super shot source set to **Plays: home** or **away**, or a Hot streaks source set to one streak, plays once each time it is turned on. Don't keep a manual source and the automatic one on together, or it plays twice.
- **The Clash.** A pre-game animation, played from its key or as a step in the walk-out sequence, with the two teams, their logos and club colours.
- **Lower thirds.** The person lower third's name box on the Output page suggests the competition's athletes; type a guest's name and title for a commentator or an official. Turn it on when they appear and off a few seconds later.
- **Stats during play.** The ladder lower third, shooting stats, last 10 goals and match up are all Source keys. Each loads fresh when shown, so it has the latest numbers.

4.6 Finals

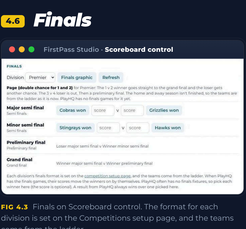


FIG 4.3 Finals on Scoreboard control. The format for each division is set on the Competitions setup page, and the teams come from the ladder.

PlayHQ often has no finals fixtures until the day. In **Finals** on Scoreboard control, pick the division, then pick each game's winner (the score is optional) as the finals are played; the finals graphic and the ladder's finals key move on with each pick. A result from PlayHQ, when it has one, always wins over one picked here.

CHAPTER 4 OF 11 — GRAPHICS

04 The graphics

Every graphic Studio draws, grouped by when it is used. Each one is added from the same **Add page** on the Output page, and its options appear as fields under its entry.

Jump to: Always on · Before the game · During play · After the game · Half-time analysis · Pre-match · Pre-competition · Pre-fixtures · Checking the outputs

All of them follow the game night, game and count picked on Scoreboard control, take the competition's colours, font and logos from its setup, and bring their teams and players in from the Squads you've built and stored once to fit when they have more in them than fits. Every page, with its link and a Preview button, is also listed on the Docs tab.

4.1 Always on

4.1.1 Scoreboard. Built, both teams with their logos and club colours, the score, the quarter and the clock, ranked or unranked, and count position (top-centre, bottom-left and so on), upper and tab on or off, and the naming rights sponsor's line.

4.1.2 Score Bug. A smaller form with the team logos and club colours, the score, the quarter and the clock, ranked or unranked, and count position (top-centre, bottom-left and so on), upper and tab on or off, and the naming rights sponsor's line.

4.1.3 Title. The competition, round and venue as a lower-third, with the venue's line above either one of the competition logo above it, tab or broken, and the naming rights sponsor under it.

4.1.4 Viewers. A watching! from the competitions YouTube live figure, with the viewer count and the Live badge. Appears the competition's connection. **Shows peak** makes it a card at the end of the night.

4.1.5 Subscribe Lower. A call to subscribe as you count, with a name and image. After the last game it becomes the end-of-night card.

4.2 Before the game

4.2.1 Countdown. Like Starts on the same pattern with a countdown to the live teams, Options heading, the words at zero, a Date over the last seconds, position and a sponsor.

4.2.2 Match up. Each team's name and logo, with Options for name, form and key players, and Name and Head to head.

4.2.3 Position match ups. All three centre, position by position, with the Options for name, form and key players, and Name and Head to head. Each player's name and team, with a list of the starting line, and per side of the end.

4.2.4 Team List. The starting five up from Warmups with squads, added if available, with the Options for name, form and key players, and Name and Head to head. Each player's name and team, with a list of the starting line, and per side of the end.

4.2.5 Intro and outro. Introduction, about 8 seconds, with the Options for name, form and key players, and Name and Head to head. Each player's name and team, with a list of the starting line, and per side of the end.

4.3 During play

4.3.1 Super shot. Automatic, 3 seconds, the scoring name's surname, Options for name, form and key players, and **Flips forms / away** the team's name in manual one.

4.3.2 On the siren. A graphic that appears during play, with Options for name, form and key players, and Name and Head to head. Each player's name and team, with a list of the starting line, and per side of the end.

4.3.3 Heating up. Shows a player's name and a list of the starting line, and per side of the end. Each player's name and team, with a list of the starting line, and per side of the end.

4.3.4 Hot streaks and moments. Name, top up or this, put the spot, the first, all season, Lights streak, on the shot, save in mark. Automatic from the goal rules, of **Flips** the scoring name.

4.3.5 Lower third. Name, Options for name, form and key players, and Name and Head to head. Each player's name and team, with a list of the starting line, and per side of the end.

4.3.6 Shooting stats. Goals, super shots and attempts per shooter from Warmups, with Options for name, form and key players, and Name and Head to head. Each player's name and team, with a list of the starting line, and per side of the end.

4.3.7 Last 10 goals. The game's most recent goals added to reveal with a Options for name, form and key players, and Name and Head to head. Each player's name and team, with a list of the starting line, and per side of the end.

4.4 After the game

4.4.1 Winner. Automatic, a winning logo, final score and, in grand slam, ended out from the stand each time it comes on. Options for a result, the crest, the live under the score, animate away.

4.4.2 Full time. Like the score from the board with each top shooter and shooting accuracy from Playing game days. **Heading** makes it a half-time card.

4.4.3 Score wins. The straight-through this game as a title. Shown to beat a fixture. Options showing the club colours of the competitors.

4.4.4 Game wins. As shown by scarlet, recorded by the Options for name, form and key players, and Name and Head to head. Each player's name and team, with a list of the starting line, and per side of the end.

4.4.5 Thanks. Thanks for watching, with the Options for name, form and key players, and Name and Head to head. Each player's name and team, with a list of the starting line, and per side of the end.

4.4.6 End of night. Shown after the last game thanks for watching. Options for name, form and key players, and Name and Head to head. Each player's name and team, with a list of the starting line, and per side of the end.

> **Cards sit in the space between the title and the scoreboard and shrink to fit when they have more in them than fits.**
> CHAPTER 4 · THE GRAPHICS

4.5 The competition

4.5.1 Leader. The competition's table with the finals places marked and every side's fixtures. Updates only, logo on an AFL light or end-made logo.

4.5.2 Fixtures Round (Full Page). The competition's table with the finals places marked and every side's fixtures. Updates only, logo on an AFL light or end-made logo.

4.5.3 Round Fixture. The competition's fixtures across all divisions with the Options for name, form and key players, and Name and Head to head. When divisions are on different rounds, each game shows its own round.

4.5.4 Tonight's Fixtures. The competition's fixtures across all divisions with the Options for name, form and key players, and Name and Head to head. Each player's name and team, with a list of the starting line, and per side of the end. Tonight's Fixtures. The streamed court has a name and a logo. Options for name, form and key players, and Name and Head to head. Each player's name and team, with a list of the starting line, and per side of the end.

4.5.5 Round Results. The game's scores for the round, with Options for name, form and key players, and Name and Head to head. Each player's name and team, with a list of the starting line, and per side of the end.

4.5.6 Top shooters. The season's leading shooters, with Options for name, form and key players, and Name and Head to head. Each player's name and team, with a list of the starting line, and per side of the end.

4.5.7 Results. The game's scores for the round, with Options for name, form and key players, and Name and Head to head. Each player's name and team, with a list of the starting line, and per side of the end.

4.5.8 Semi-finals. The semi-finals pairings and their scores, with Options for name, form and key players, and Name and Head to head. Each player's name and team, with a list of the starting line, and per side of the end.

4.5.9 Championship. The grand final fixture, with Options for name, form and key players, and Name and Head to head. Each player's name and team, with a list of the starting line, and per side of the end.

4.5.10 Others. The competition's fixtures, results or ladder for each other division, with Options for name, form and key players, and Name and Head to head. Each player's name and team, with a list of the starting line, and per side of the end.

4.6 The breaks

4.6.1 Sponsor card. One approved image, tagline, offer and a QR code to their website, taking turns by the Options you set each interval, in order.

4.6.2 Sponsor loop. One approved image, tagline, offer and a QR code to their website, taking turns by the Options you set each interval, in order.

4.6.3 PIVT and FirstPass Studio ID. Automatic, a PIVT and FirstPass Studio ID for a break with nothing else. Options for name, form and key players, and Name and Head to head. Each player's name and team, with a list of the starting line, and per side of the end.

4.7 Checking the outputs

4.7.1 Test pattern. Bars, fine lines, ramps and a still. Automatic, a test pattern for checking a capture device and the levels and tones before the stream. Nothing else.

CHAPTER 6 OF 11

Competitions setup

Everything about a competition that doesn't come from PlayHQ lives on one page: its PlayHQ link, the names on the graphics, the game rules, finals, game day, colours, font and logos. Set it up once a season and every graphic follows.

PlayHQ · Names on the graphics · Game rules · Finals · Game day · Colours and font · Logos · Live viewers · Templates, export and more

Studio keeps a setup per competition. The one marked **In use** is what every graphic, Scoreboard control, the Sponsors page and the Stream Deck use. Changes aren't kept until you press **Save** (or **Save and use this competition**), and leaving the page with unsaved changes asks first.



FIG 6.1 The competitions list. Start from a template file (the repository has Thunder League, AMHNA Nationals, GIO WANL and West Coast Thunder) or press New competition for Studio's defaults.

6.1 PlayHQ

Paste the address of the organisation's page on playhq.com and press **Find its competitions**; then pick the competition and the season. For a carnival or a league run under another body, use the organisation the competition is listed under. Nothing else about PlayHQ is needed: no account, no key.

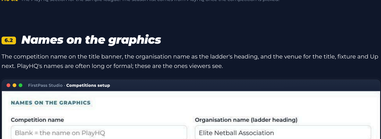


FIG 6.2 The PlayHQ section for the sample league. The season list comes from PlayHQ once the competition is picked.

6.2 Names on the graphics

The competition name on the title banner, the organisation name as the ladder's heading, and the venue for the title, fixture and Up next. PlayHQ's names are often long or formal; these are the ones viewers see.



FIG 6.3 Names on the graphics.

6.3 Game rules

Quarters or halves and their length set the clock, the quarter buttons and the wording of the break cards. **Super shots** turns on the +2 scoring, the celebration, the super shot tab in the last minutes of each quarter and the Super columns on the shooting graphics; a division can have its own setting, by its name or by a word in it ("Open", "Men"). The break lengths let the sequence editor say how many loops of a break sequence fit in the break.



FIG 6.4 Game rules. The sample league plays 15-minute quarters with super shots in the last 5 minutes, except in its Under 21 division.

6.4 Finals

Pick the finals format for the competition, or per division: it decides the finals places the ladder marks (double chance or a week off, elimination, playing for third) and the bracket the finals graphic draws week by week. PlayHQ's finals games are used when it has them; otherwise the winners are picked on Scoreboard control.

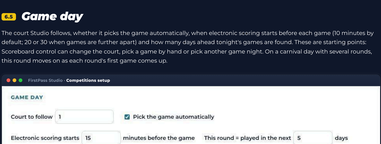


FIG 6.5 Finals formats, per division where they differ.

6.5 Game day

The court Studio follows, whether it picks the game automatically, when electronic scoring starts before each game (10 minutes by default; 20 or 30 when games are further apart) and how many days ahead tonight's games are found. These are starting points: Scoreboard control can change the court, pick a game by hand or pick another game night. On a carnival day with several rounds, this round moves on as each round's first game comes up.



FIG 6.6 Game day.

6.6 Colours and font



FIG 6.7 Colours and font, with the preview.

The competition's colours, see-through dark panels, and the font. Paste a brand kit's text into **Brand kit colours** and every HEX or RGB value in it becomes a swatch in each colour picker. Add a font file (TTF, OTF, WOFF or WOFF2, one per weight or a variable font) and it travels with the competition and its template, so the graphics look the same on any Mac; every font is sized to match the capitals of Montserrat, which the graphics were drawn with. The preview under the pickers shows a rough title and scoreboard in the colours before you save.

6.7 Logos

The **competition logo** has a light mode and a dark mode version (one is used for both if that's all you add); it shows beside the title in the branding logos, above the title with its option on, in the corner of the ladder and in the logo animations. **Team logos** are matched to PlayHQ team names by file name, so drop in a folder of files named after the teams. Logos and colours belong to clubs: every team of a club, in every division, gets the club's logo and colour, worked out from the team name without its division ("Falcons 2Yo" is Falcons). Fix a wrong match in the **Club** box. A club colour goes behind its teams' names on the scoreboard and behind their logo on The Clash; without one the competition's colours are used.



FIG 6.8 Competition logo, light and dark mode.

FIG 6.9 Team logos and the teams on PlayHQ, grouped by club with a club colour each.

> **Logos and colours belong to clubs: every team of a club, in every division, gets the club's logo and colour.**
>
> CHAPTER 6 · COMPETITIONS SETUP

6.8 Live viewers

Add the YouTube channels and Facebook Pages the competition streams to. While a video output is on, Studio adds up how many are watching every 30 seconds, for the Output page, the viewers graphic (which shows only from the minimum set here) and the viewer-minutes in the sponsor report. The page explains how to get a YouTube API key and a Facebook Page token; both are kept on this Mac only, never in iCloud, templates or exports.

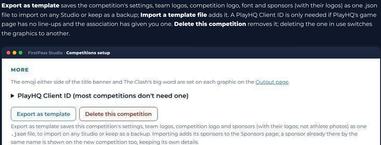


FIG 6.10 Live viewers.

6.9 Templates, export and more

Export as template saves the competition's settings, team logos, competition logo, font and sponsors (with their logos) as one .json file to import on any Studio or keep as a backup; **import a template file** adds it. A PlayHQ Client ID is only needed if PlayHQ's game page has no line-ups and the association has given you one. **Delete this competition** removes it; deleting the one in use switches the graphics to another.

FIG 6.11 The More section: Client ID, export and delete.

CHAPTER 7 OF 11

Sponsors

The Sponsors tab turns sponsorship into something you can sell, run and prove: a tiered library of sponsors, spots on the graphics that fill themselves, ad breaks that take turns fairly, and a report of what each sponsor got.

1 Tiers · 2 A sponsor · 3 Spots and moments · 4 Proof of play



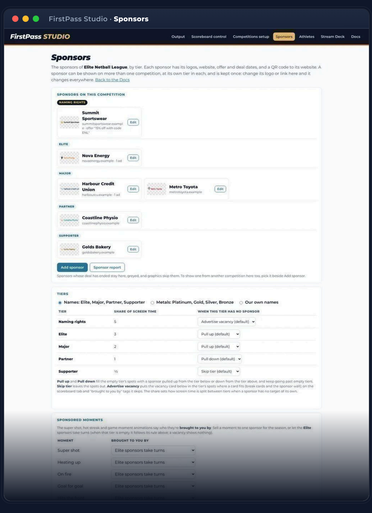


FIG 7.1

The Sponsors page for the sample league: sponsors by tier with tonight's exposure meter, the tier rules, sponsored moments and the vacancy card.

7.1 Tiers

Five fixed ranks, named per competition: the **Names** preset (Naming rights, Elite, Major, Partner, Supporter), the **Metals** preset (Naming rights, Platinum, Gold, Silver, Bronze) or your own. Each tier's **share of screen time** decides how turns are split when a sponsor has no target of its own. Each tier gets different places:

TIER	WHERE IT SHOWS
Naming rights	A tab on the scoreboard, under the title banner, first on the sponsor wall, and everything below.
Elite	The sponsored moments (super shot, streaks), ads in the breaks, the break card, and everything below.
Major	Ads and break cards, "presented by" footers on the stats cards, countdown and top shooters.
Partner	A slot in the score ticker and the logo spots.
Supporter	The sponsor wall and the end-of-night strip.

When this tier has no sponsor says what a spot does when its tier is empty: **Pull up** fills it from the tier below, **Pull down** from the tier above, **Skip tier** leaves it out, and **Advertise vacancy** shows the competition's vacancy card wherever a card fits. The vacancy card has a heading, a contact, a message and a link or email for its QR code, so an empty tier finds its own sponsor.

> **The vacancy card has a heading, a contact, a message and a link or email for its QR code, so an empty tier finds its own sponsor.**
>
> CHAPTER 7 · SPONSORS

7.2 A sponsor

A sponsor has a name, tagline, website, offer, brand colour, category, deal dates, and optional targets per stream (minutes on screen, ad plays). Category keeps competing sponsors apart in rotation. Add a colour logo and a white one for dark panels (without a white one, the colour logo is made white). A sponsor is kept once and can be **shown on** several competitions at its own tier in each; sponsors whose deal has ended stay greyed and the graphics skip them.

Ads are videos or stills (up to 10 per sponsor) in the media folder. A **Sponsor ads** step in a sequence plays the next ads due, taking turns fairly by tier, each video to its end with sound and each still for its seconds; tick **House ad** and the PIVT and FirstPass Studio ad plays when nothing is due. The **QR code** points to the sponsor's website with tracking tags, so visits from the stream show in their analytics; save it as a PNG for the sponsor's own use.

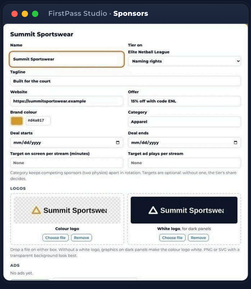


FIG 7.2 The sponsor editor: details, logos, ads, the QR code with tracking tags, and the competitions it is shown on.

7.3 Spots and moments

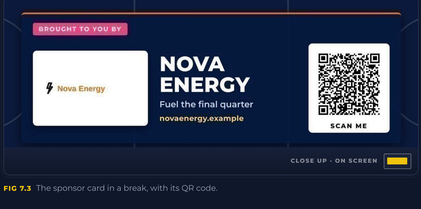


FIG 7.3 The sponsor card in a break, with its QR code.

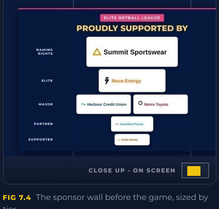


FIG 7.4 The sponsor wall before the game, sized by tier.

Every spot picks its sponsor from the Sponsors page by default. Each graphic's **Sponsor** option can pin one sponsor or turn the spot off. **Sponsored moments** sells the super shot, each hot streak and each game moment to one sponsor for the season, whose name appears in a "brought to you by" tag on the animation; by default the Elite sponsors take turns.

7.4 Proof of play

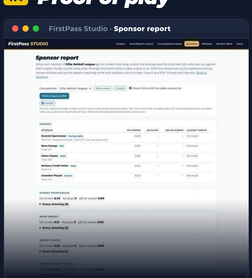


FIG 7.5 The sponsor report for the sample night.

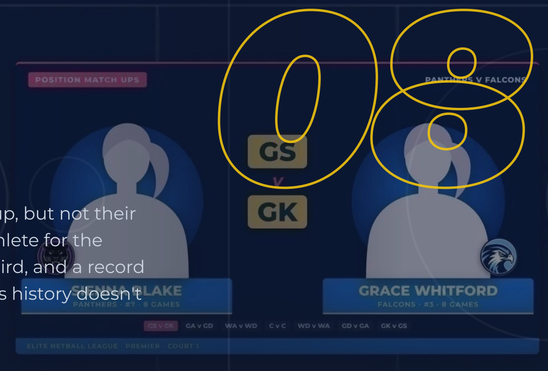
While a video output is on, Studio logs every second each sponsor is on screen and every ad played, matched to the scene's sources, and reads the live viewer count every 30 seconds. **Sponsor report** turns that into a page per night (or a run of nights) with minutes on screen, ad plays, viewer-minutes, peak and average viewers per sponsor, ready to print or save as a PDF for the sponsor. A tick box counts time with no video output on, for a rehearsal. With more than one output, only Output 1 and outputs with **Counts for sponsors** ticked are counted, and each counts as on air only while it's on: an output set to **Web links only** is off.

CHAPTER 8 OF 11

Athletes

PlayHQ gives Studio every athlete in a line-up, but not their faces. The Athletes tab holds a photo per athlete for the position match ups and the person lower third, and a record of positions and games played that PlayHQ's history doesn't cover.

1. Photos
2. Positions and games played



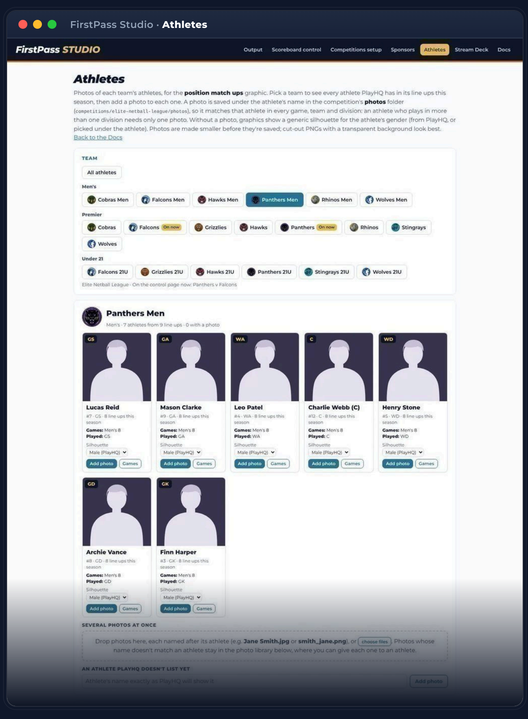


FIG 8.1

The Athletes page: teams grouped by division, every athlete PlayHQ has in a line-up this season, and the photo library.

8.1 Photos

Pick a team and add a photo to each athlete, or drop a folder of photos named after the athletes ("Jane Smith.jpg" or "smith_jane.png") on **Several photos at once**; any that don't match wait in the photo library to be given to someone. A photo is saved under the athlete's name for the competition, so an athlete in two divisions needs one photo. Studio also remembers each athlete's PlayHQ id, so if PlayHQ changes a player's name during the season, their photo still finds them. Photos are made smaller when saved; cut-out PNGs with a transparent background look best. Without a photo the graphics show a silhouette for the athlete's gender, taken from the division (men's, women's; mixed and open are neutral) and changeable under the athlete. **An athlete PlayHQ doesn't list yet** adds one ahead of their first game.

A photo only goes on the stream and the Socials pictures once **Approved for photo use** is ticked under the athlete. Tick it when the athlete, or their parent or guardian, has agreed to their image being used. Until then the graphics show the silhouette. **Approve all for photo use** ticks every athlete listed for the team at once.

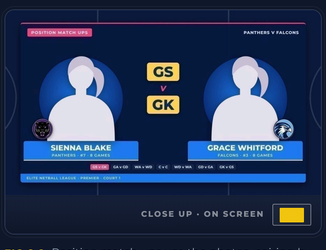


FIG 8.2 Position match ups use the photos, pairing by pairing.



FIG 8.3 The athlete lower third, named from the register as you type on the Output page.

> ***A photo is saved under the athlete's name for the competition, so an athlete in two divisions needs one photo.***

CHAPTER 8 · ATHLETES

8.2 Positions and games played

Games under an athlete opens their record. Positions played in a PlayHQ line-up this season are ticked already; tick any others from other seasons or competitions. Games played takes a line per division (and club): the number as of a date, to which Studio adds the games PlayHQ shows after it, so milestones keep up through the season. With no line, a division's games are this season's from PlayHQ.

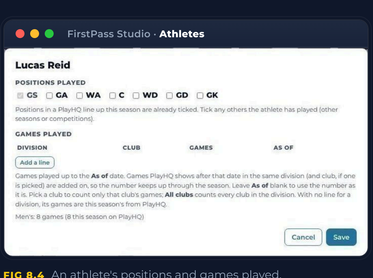
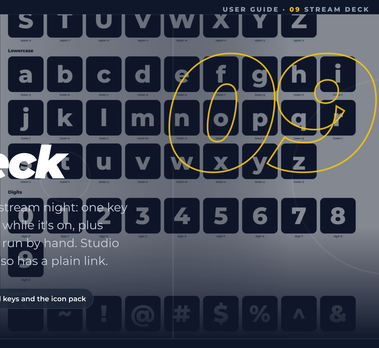


FIG 8.4 An athlete's positions and games played.



CHAPTER 9 OF 11

Stream Deck

A Stream Deck is the control surface for stream night: one key per source, scene and sequence, each lit while it's on, plus score, clock and quarter keys for a game run by hand. Studio ships its own plugin, and every control also has a plain link.

1. The plugin
2. Links for everything else
3. Timed keys and the icon pack

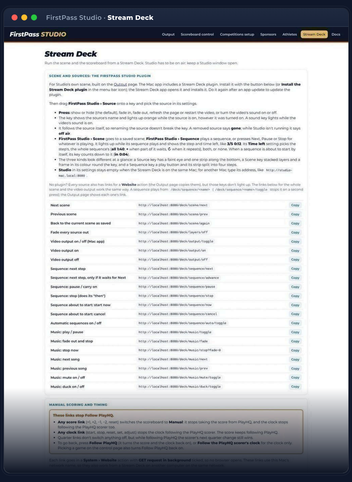


FIG 9.1

The Stream Deck tab: the plugin, the scoring and clock links, and the full reference below them.

9.1 The plugin

Install it from the 1P menu (**Install the Stream Deck plugin**) or the button on the Output page; the Stream Deck app opens and installs it. Do the same after an app update. The **Studio** field in a key's settings stays empty when the Stream Deck is on the same Mac; for another Mac, type its address, like `http://studio-mac.local:8080`. Four kinds of key, each a different shape at a glance:

KEY	LOOKS	PRESS	SHOWS
Source	A faint eye and one strip along the bottom	Show or hide (default), fade in, fade out, refresh the page or restart the video, video sound on or off	The source's name; lit in its output's colour while it's on, however it was turned on
Scene	Stacked layers and a frame in its colour	Go to a scene, next or previous scene, or back to the current scene as saved	Lit while its scene is on, with a yellow dot when sources have changed since
Sequence	A play button and a strip in four steps	Play or stop a sequence, play from the first step, or Next, Next when waiting, Pause and Stop for whatever is playing	Step and time left, like **3/5 0:12**, the whole sequence's time, both or none; **in 0:04** when one is about to start by itself
Switcher	A camera and a strip in two halves	A camera to preview (default) or program, stinger to a camera, Cut, Auto, an upstream or downstream key, Fade to black, run a macro	The camera's name from the switcher; red on program and green on preview, a key or Fade to black red while on air; a yellow dot while the switcher isn't answering

Keys follow their source, scene or sequence by id, so renaming on the Output page doesn't break them; a removed one says **gone**, and while Studio is off air they say **off air**. Colour groups keys by the part of the stream: green pre-game, blue live, purple quarter time, grey post-game, yellow ads, or the scene's or sequence's own colour.

With more than one output, each key has an **Output** setting: the output it shows a source on, plays a scene on or plays a sequence on (a Switcher key has none: there is one switcher). A lit key takes its output's colour (Output 1 orange, 2 pink, 3 blue, 4 green) with a black title, so a commentators' Stream Deck of Commentary keys lights pink while the stream's keys light orange. A key for a removed output says **no output**.

> ***Keys follow their source, scene or sequence by id, so renaming on the Output page doesn't break them.***
>
> CHAPTER 9 · STREAM DECK

9.2 Links for everything else

Every control is also a link, for a **System > Website** action (tick **GET request in background**), a Multi Action, or any program on the network. Scoreboard control, the Output page and the Stream Deck tab copy them, with this Mac's Stream Deck key after the port (`…:8080/k/key/deck/…`) so no web page can press them. The Studio plugin needs no key: the first time you press one of its keys the Mac asks you to **Allow** it, and from then on it signs every press with its own key pair. A link without it is refused, so copy any older Website keys' links again from Studio. With more than one Stream Deck, give each its own link set: pick **New link set…** in **Copy links for** on the Stream Deck tab, name it after the Stream Deck, and copy its links from there. If a link gets out, choose **Devices…** in the menu bar icon and press **Make a new key** on that set: its links stop working and you copy them again, while the other Stream Decks carry on. A program on another computer can ask for a key of its own: the Mac asks you to **Allow** or **Deny** it by name and address, and Devices… can take that one away without touching your other keys. The useful ones:

GROUP	LINKS
Scores	Home and away +1, +2, −1, −2; reset to 0; play a super shot or any streak without changing the score.
Scoring mode	Follow PlayHQ, Manual, toggle. Any score key switches to Manual.
Clock	Start, stop, toggle, reset, set to a time, add or take off any amount, follow the PlayHQ clock on or off, quarter length.
Quarter	Q1 to Q4, extra time, next, previous. The clock is separate; pair them in a Multi Action.
Game	Automatic selection on, off, toggle; the court to follow.
Ticker	Faster, slower, normal, a set speed.
Sources	Show, hide, toggle, restart, video sound; every source out; video output on or off.
Scenes	Go to a scene, next, previous, back to the scene as saved.
Sequences	Play, toggle, next, advance (only when waiting), pause, stop, start the one about to start now, cancel it, automatic sequences on or off.
Music	Play, pause, resume, stop, fade out, next, previous, load a song or playlist, mute, duck; add `?fade=2` for a 2-second fade.
Sponsors	Play the next sponsor ad due.
Switcher	Cut, auto, stinger (to a camera or what's on preview), fade to black, a camera to preview or program, an upstream or downstream key on, off or toggled, run a macro.

A LAYOUT THAT WORKS

Put the scene keys in a row across the top (Pre-game, Live game, Clear), the sequence keys under them with Next when waiting and Stop at the end, and a page of Source keys for the lower thirds and stats. A game run by hand gets a second page: home scores left, away scores right, the clock and quarter keys down the middle.

9.3 Timed keys and the icon pack

A second plugin, **Timed Sequence**, arms a key for a time of day and then presses a list of links with delays between them: reset the board at 18:55 and start the clock at 19:00, say. It is built from the repository. The **FirstPass Studio Characters** icon pack gives Stream Deck keys letters, numbers and symbols in the Studio style, for labelling keys by hand.

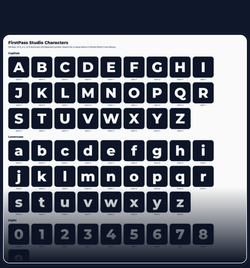


FIG 9.2 The Characters icon pack: A to Z, a to z, 0 to 9 and the usual symbols on navy.



CHAPTER 10 OF 11

Settings and care

Where Studio keeps its settings, how to carry them between Macs and back them up, how updates arrive, and what to do when something isn't right.

1 The 1P menu
2 The data folder
3 Settings in iCloud Drive
4 Backup and restore
5 Updates
6 Links and other devices
7 When something isn't right
8 Data and bandwidth

10.1 The 1P menu

The menu bar icon is Studio's home. From top to bottom: the version and whether Studio is on air; **Stop Studio** (while on air); **Open Studio**, Scoreboard control, Competition setup and **Copy the Studio link**; **Devices…**, **Install the Stream Deck plugin**; **Open the settings folder**, **Settings folder** (iCloud), **Back up…** and **Restore…**; **Open the debug log**, **Open the app log**, **Restart Studio**, **Check for updates**; **Open at login**, **Only this Mac can change things**, **What's new…** and Quit.

10.2 The data folder

Everything Studio saves lives in one folder outside the app, `~/Library/Application Support/FirstPass Studio`, so updates never touch it. In it: **competitions/** (a folder per competition with its settings, logos, font, athlete photos and register), **sponsors/**, the scene, sequences, music and playlists, the **media** folder, **control.json** (Scoreboard control's choices), **scoreboard-state.json** (the game, scores and clock, saved every few seconds so a restart mid-game loses nothing), **config.json** for this Mac's own settings, **logs/** and **backups/**.

10.3 Settings in iCloud Drive

Settings folder > Move to iCloud Drive… moves the settings into a folder you choose in iCloud Drive, in a subfolder named after this Mac, so each Mac on the same Apple account keeps its own and none saves over another's. The debug log, scores and clock, caches and window positions always stay on the Mac. **Restore from another Mac…** copies another Mac's settings over this one's (competitions, scenes, sequences, music, control choices), backing this Mac's up first. **Move back to this Mac…** undoes the move. Studio goes off air for a moment for each and comes straight back.

10.4 Backup and restore

Back up… saves every setting in one zip named with the date and time, with notes you type and, if you choose, the media folder; beside it goes a text summary listing the competitions, sequences, playlists, sources, scenes and sponsors inside. **Restore…** shows when and where a backup was made and what it holds, backs up this Mac's settings first, and replaces them only once the whole zip has unpacked and checked out. This Mac keeps its own **Only this Mac can change things** choice, port and link name whatever the backup says. Make a backup after setting up a season and before trying anything new on the Output page.

10.5 Updates

Updates download by themselves and install when the app quits, never mid-stream; **Check for updates** looks now, and **Restart to update** appears when one is ready. If a newer version has come out since that one downloaded, it asks whether to get the newer one first or install the one it named. The first launch after an update shows **What's new**, the changes since the version you last opened. When an update changes how settings are saved, Studio backs the data folder up to **backups/** first; settings saved by a newer Studio open read-only in an older one.

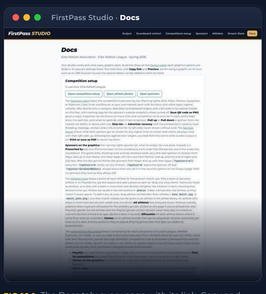


FIG 10.1 The Docs tab; every page with its link, Copy and Preview, the page options, and the notes this guide draws on.

> **Updates download by themselves and install when the app quits, never mid-stream.**
>
> CHAPTER 10 · SETTINGS AND CARE

10.6 Links and other devices

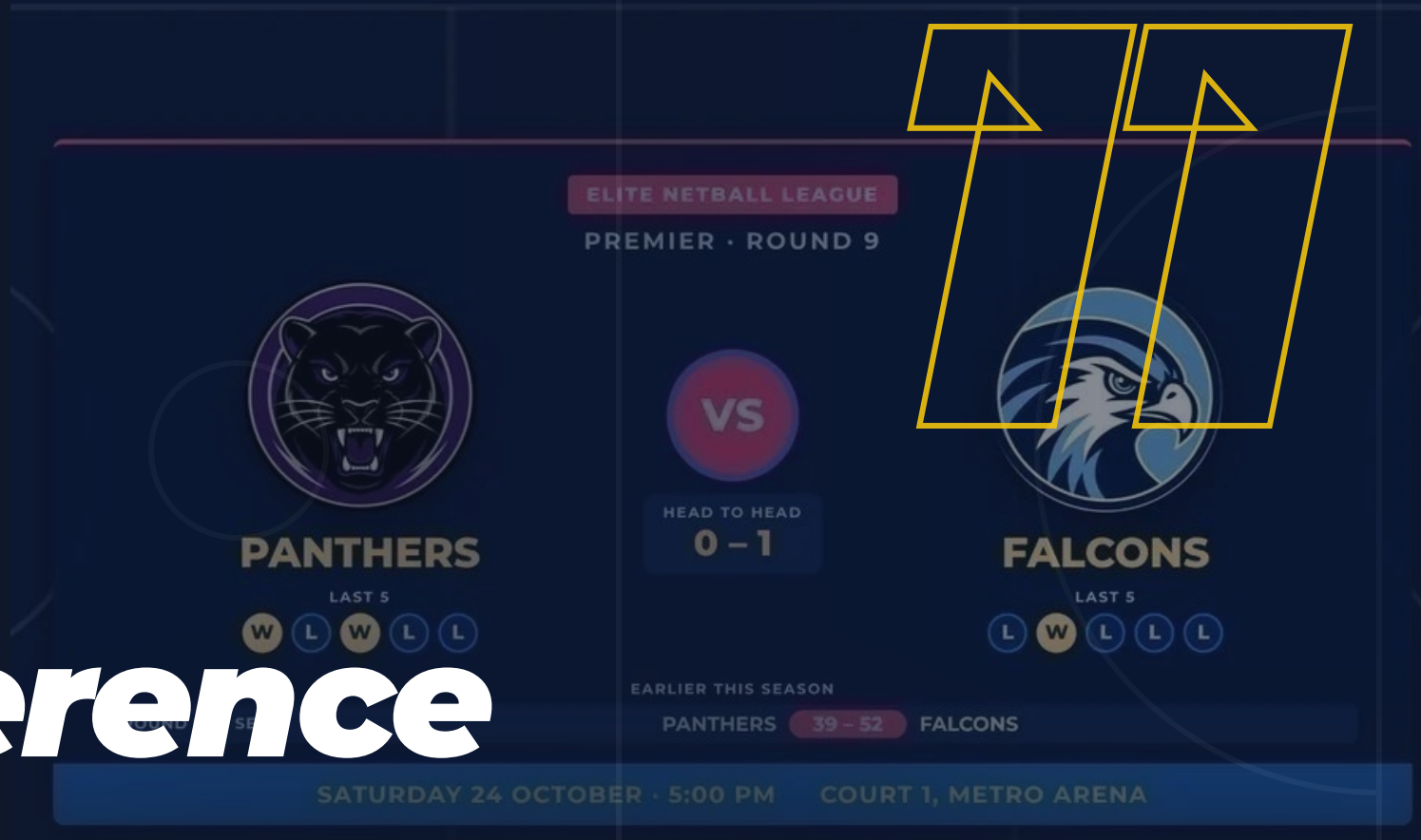
Studio's links use this Mac's Bonjour name (the Local hostname in System Settings > General > Sharing), so the same link works from a phone, another computer or a Stream Deck on the network; `localhost` still works on the Mac. Anything on the network can open the graphics. Only this Mac can change scores, the scene, sequences and settings: **Only this Mac can change things** is ticked every time Studio opens. To score from a phone or let another computer drive, pair it instead: open `/pair.html` on it (the Studio link with `/pair.html` on the end), type a name and press **Ask the Mac**, then choose **Allow scoring only** or **Allow everything** on the Mac. A paired device works with the box ticked, and **Devices…** in the menu bar icon lists it, changes what it may do, or removes it. Unticking the box lets every device on the network in until Studio quits; keep that for emergencies. On a network you don't control, leave it ticked, or block incoming connections to FirstPass Studio in the Mac's firewall.

10.7 When something isn't right

WHAT YOU SEE	WHAT TO DO
The board doesn't move before the game	Normal: nothing is read before the electronic scoring window opens. Scoreboard control says when it will start checking.
The board stops mid-game	PlayHQ isn't answering. Switch to Manual and run it by hand; Follow PlayHQ picks it up again when PlayHQ is back.
The wrong night or game is on	Pick the game night on Scoreboard control and pick the game; tick automatic selection again afterwards.
A key says off air	No Studio window is open. Open Studio from the 1P menu.
A graphic looks stale	Press its Restart key or turn it off and on; with **Load the page fresh** on it reloads each time.
Studio itself stops	**Restart Studio** in the 1P menu brings it back with the scores and clock it had.
Anything else	**Open the debug log** in the 1P menu. Everything Studio does and every PlayHQ answer is in it, 15 MB at most. Send it with a note of the time.

10.8 Data and bandwidth

The graphics only ever talk to Studio on the Mac. Studio alone contacts PlayHQ: one small request every 5 seconds for the followed game (or as set, down to one a second), the ticker's other courts every 30 seconds, the fixture about once a minute for automatic selection, and the rest once when a graphic loads. The Production Planner and the Socials tab read the same fixture, results and ladders through the same cache, and finished games' line ups once, so they add nothing; the Fixture board reads other courts' live scores the ticker's way, so it adds nothing while the ticker is on and at most one request a minute per live game when it isn't. Commentator stats' auto-refresh asks Studio only; its team lists are read at the start and at each quarter break. Finished games are read once and kept. Extra outputs read the same data from Studio, so they add nothing here. It all comes to well under a megabyte an hour, so a phone hotspot is enough for Studio; the stream itself is the bandwidth.



CHAPTER 11 OF 11

Quick reference

1 Keyboard
2 Stream night in nine lines
3 Where things are set

11.1 Keyboard

WHERE	KEYS
Scoreboard control	A / Z home +1 / −1, K / M away +1 / −1, Shift for +2 / −2, Space starts or stops the clock.
Any Studio window	⌘1 to ⌘9 and ⌘0 go to Output, Production Planner, Scoreboard control, Competitions setup, Sponsors, Athletes, Stream Deck, Socials, Recordings, Docs. ⌘N new window, ⇧⌘O open in the browser, ⇧⌘C copy the page link, ⌘[back, ⌥⌘W close every Studio window.

11.2 Stream night in nine lines

01
Open Studio from the 1P menu and keep the Output page open all night.

02
Check the competition in use and the court on Competitions setup.

03
On Scoreboard control, the first game on your court is picked with Follow PlayHQ on.

04
Turn on the video output your switcher takes and show one graphic to see it land.

05
Play a few seconds of music and an animation with sound.

06
Press a Source key and watch the Output page follow it; put scoring back to Follow PlayHQ after testing.

07
Go to the Live game scene and let the sequences run the breaks.

08
Pick finals winners as they happen; turn on lower thirds when people appear.

09
Press Stream close, turn the output off, and send the sponsor report.

11.3 Where things are set

TO CHANGE	GO TO
Colours, font, logos, rules, finals format, court	Competitions setup
Which game, scoring mode, score, clock, finals winners	Scoreboard control
What's on screen, scenes, sequences, playlists, music, mics, outputs	Output
A graphic's options (heading, position, division, animate away)	Output, under the source's address
Sponsors, tiers, ads, moments, the report	Sponsors
Photos, positions, games played	Athletes
Each night's court, the commentators and who calls each game	Production Planner
Keys and links	Stream Deck
Pictures of a game night for social media	Socials
Backups, iCloud, updates, the debug log	The 1P menu

FirstPass Studio is made by PIVT. The Elite Netball League, its clubs, athletes and sponsors are fictional.

FirstPass STUDIO

MADE BY PIVT

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